

Problem-3

$$x = [6.7, 3.1, 5.6, 2.4]$$

$$w_1 = 1, w_2 = 2, w_3 = 3, w_4 = 4$$

$$b = -2$$

formula

$$Z = x_1 w_1 + x_2 w_2 + x_3 w_3 + x_4 w_4 - b$$

Sol:

$$Z = (6.7 \times 1) + (3.1 \times 2) + (5.6 \times 3) + (2.4 \times 4) - 2$$

$$= 6.7 + 6.2 + 16.8 + 9.6 - 2$$

$$= 39.3 - 2$$

$$Z = \underline{\underline{37.3}}$$

Sigmoid

$$\sigma(37.3) = \frac{1}{1 + e^{-z}}$$

$$= \frac{1}{1 + e^{-37.3}} = 1$$

$$h = 1$$

Output layer

Setosa

$$z_{\text{setosa}} = 1(1.8) + 0.1$$

$$z_{\text{setosa}} = 1.9$$

$$\sigma(1.9) = 0.886$$

versicolor

$$z_{\text{versicolor}} = 1(1.6) + 0.2$$

$$= 1.8$$

$$\sigma(1.8) = 0.858$$

virginica

$$z_{\text{virginica}} = 1(1.9) + 0.15$$

$$z_{\text{virginica}} = 2.05$$

$$\sigma(2.05) = \underline{\underline{0.886}}$$